Criterio de la integral ☹

Si la integral converge (Da un numero), o diverge (da ∞), la serie converge o diverge

1) Ver si es decreciente, $f'(x)$

2) Resolver la integral definida y ver si converge o diverge

$$\sum_{n=p}^{\infty} f(x) \rightarrow \int_{p}^{+\infty} f(x)$$

$$\lim_{n \to +\infty} f(x) - f(p)$$

1) $$\sum_{n=0}^{\infty} n \cdot e^{1-n^2}$$

$$f(x) = x \cdot e^{1-x^2}$$

Es decreciente?

$$[f(x)g(x)]' = f'(x)g(x) + f(x)g'(x)$$

$$[e^{f(x)}]' = e^{f(x)} f'(x)$$

$$f(x) = x \cdot e^{1-x^2}$$

$$x' = 1$$

$$e^{1-x^2} + x e^{1-x^2} \cdot -2x$$

$$e^{1-x^2} - 2x^2 e^{1-x^2} = 0$$

$$e^{1-x^2}(1 - 2x^2) = 0$$

$$\cancel{e^{1-x^2} = 0}$$

$$1 - 2x^2 = 0$$

$$2x^2 = 1$$

$$x^2 = \frac{1}{2} \rightarrow x = \pm\sqrt{\frac{1}{2}} \rightarrow \sqrt{\frac{1}{2}}$$

En series se toma el > 0

$(1-2x^2)$, $x = \sqrt{\frac{1}{2}}$

x	$-$	$\bullet$	$+$
$-x$	$+$	$\circ$	$-$

	$-\infty$	$\sqrt{\frac{1}{2}}$	$+\infty$
$1-2x^2$	$+$	$\bullet$	$-$
$f'(x)$	$+$		$-$
$f(x)$	$\nearrow$		$\searrow$

Decreciente

$$\sum_{n=0}^{\infty} n \cdot e^{1-n^2}$$

$$\int_0^{+\infty} x \cdot e^{1-x^2} \, dx$$

$$\int x \cdot e^{1-x^2} \, dx$$

$u = 1-x^2$
$du = -2x$
$-\frac{1}{2} du = x \, dx$

$$\int e^u \cdot -\frac{1}{2} \cdot du$$

2. $\int k f(x)\,dx = k\int f(x)\,dx$

6. $\int e^x dx = e^x + c$

$$-\frac{1}{2} \int e^u \, du$$

$$-\frac{1}{2} e^u \longrightarrow -\frac{1}{2} \cdot e^{1-x^2}$$

$$+\infty \leftarrow \lim_{x\to+\infty} f(x)$$

$$\frac{-1}{2}\cdot e^{1-x^2}$$

$+\infty$

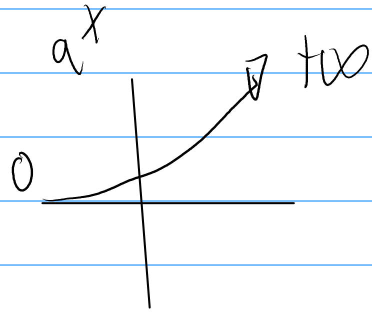
$0 \to f(k),\ k=0$

$$\lim_{x\to+\infty} \frac{-1}{2}\cdot e^{1-x^2} - \frac{-1}{2}\cdot e^{1-0^2}$$

$$\int_a^b f(x) = F(b) - F(a)$$

$$\lim_{h\to+\infty} f(h) - f(a)$$

$$\frac{-1}{2}\cdot \overset{0}{\cancel{e^{-\infty}}} + \frac{1}{2}\cdot e$$



$$\frac{e}{2} \leftarrow k$$

$$\boxed{\therefore \sum_{n=0}^{\infty} n\cdot e^{1-n^2} \text{ converge}}$$

Solo hacer derivada, no tabla

2) $$\sum_{n=2}^{\infty} \frac{1}{n\ln(n)}$$

$$\left[\frac{1}{f(x)}\right]' = \frac{-1}{f^2(x)} f'(x)$$

Decreciente?

Con $f(x) = x\ln(x)$

$$f(x) = \frac{1}{x\ln(x)}$$

$$\left[\frac{1}{f(x)}\right]' = \frac{-1}{f^2(x)} f'(x) \qquad [f(x)g(x)]' = f'(x)g(x) + f(x)g'(x)$$

$$[\ln x]' = \frac{1}{x}$$

$$\frac{-1}{(x\ln(x))2} \cdot \ln(x) + \cancel{x} \cdot \frac{1}{\cancel{x}}$$

$$\frac{-1}{(x\ln(x))2} (\ln(x) + 1)$$

Como $f(x)$ es < 0 sí ES decreciente

$$\sum_{n=2}^{\infty} \frac{1}{n\ln(n)}$$

$$\int_2^{+\infty} \frac{1}{x\ln(x)}\, dx$$

$$\int_2^{+\infty} \frac{1 \to 1}{x \to \ln(x)}\, dx$$

$$\int_2^{+\infty} \frac{1 \to 1}{x \to \ln(x)}\, dx \qquad u = \ln(x), \quad du = \frac{1}{x}\,dx$$

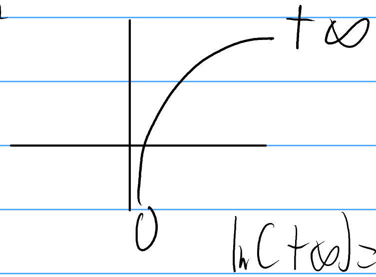
$$\int \frac{1}{u}\,du$$

1. $\int \frac{1}{x}dx = \ln|x| + c$

$$\ln|u|$$

$$\ln|\ln(x)| \Big|_2^{+\infty}$$

ln



$$\ln(+\infty) = +\infty$$

$$\lim_{x\to+\infty} \ln|\underbrace{\ln(+\infty)}_{\ln(+\infty)}| - \underbrace{\ln|\ln(2)}_{\pi}|$$

$$+\infty - \pi$$

$+\infty$ ∴ La $\int$ diverge

$$\boxed{\therefore \text{ La serie } \sum_{n=2}^{\infty} \frac{1}{n\ln(n)} \text{ diverge}}$$

$$\sum_{n=1}^{\infty} \frac{\ln(n)}{n^2}$$

Decreciente ?

$$f(x) = \frac{\ln(x)}{x^2} \quad \frac{f(x)}{g(x)} \qquad \left[\frac{f(x)}{g(x)}\right]' = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

$$\frac{\frac{1}{\cancel{x}} \cdot \cancel{x}^2 - \ln(x) \cdot 2x}{(x^2)^2}$$

$$\frac{x - 2x\ln(x)}{x^4} = 0$$

$$\frac{\cancel{x}(1 - 2\ln(x))}{\cancel{x^4}} = 0 \qquad PC$$

$$x = 0 \quad \vee \quad f(x) \nexists \text{ Para } x = 0$$

$$\frac{1 - 2\ln(x)}{x^3} = 0 \qquad 0$$

$$1 - 2\ln(x) = 0$$
$$1 = 2\ln(x)$$
$$\ln(x) = \frac{1}{2} \qquad \ln(x) = K$$
$$e^{\cancel{\ln}(x)} = e^{\frac{1}{2}} \qquad e^{\ln(x)} = e^{K}$$
$$x = e^{\frac{1}{2}} \qquad x = e^{K}$$

$$\sum_{n=1}^{\infty} \frac{\ln(n)}{n^2}$$

$$u \cdot v - \int v\, du$$

Inv. Trig.
Log.
Algeb.
Trigon.
Exponenciales

$$\int \frac{\ln(x)}{x^2}$$

Deriva

$u = \ln(x)$

$du = \frac{1}{x}\,dx$

Integra

$dv = \frac{1}{x^2}$

$v = \int \frac{1}{x^2}$

$v = \frac{-1}{x}$

$$\int \frac{1}{x^2}$$

$$\int x^{-2}$$

$$\frac{x^{-2+1}}{-2+1}$$

$$\frac{x^{-1}}{-1}$$

$$= \frac{-1}{x}$$

$$\ln(x) \cdot -\frac{1}{x} - \int \frac{-1}{x} \cdot \frac{1}{x}\, dx$$

$$\frac{-\ln(x)}{x} - - \int \frac{1}{x^2}\, dx$$

$$\frac{-\ln(x)}{x} + \int \frac{1}{x^2}\, dx$$

$$\frac{-\ln(x)}{x} - \frac{1}{x}$$

$$\left.\frac{-\ln(x)}{x} - \frac{1}{x}\right|_{1}^{+\infty}$$

$$\lim_{x \to +\infty} \frac{-\ln(x)}{x} - - \frac{1}{1}$$

$$\frac{f(x)}{g(x)}, \quad f(x) < g(x)$$

$$\ln(x) < x^n$$

$$\lim_{x \to +\infty} \cancel{\frac{-\ln(x)}{x}} + 1 \qquad = 0$$

0

1 ✓ Converge

$$\boxed{\sum_{n=1}^{\infty} \frac{\ln(n)}{n^2} \quad \text{Converge}}$$

$$\sum_{n=1}^{\infty} \frac{n}{e^{n^2}}$$

$$\left(e^{f(x)}\right)' = e^{f(x)} \cdot f'(x)$$

$$f(x) = \frac{x}{e^{x^2}}$$

$$\left[\frac{f(x)}{g(x)}\right]' = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

$$f'(x) = \frac{e^{x^2} - x e^{x^2} \cdot 2x}{e^{x^4}}$$

$$= \frac{\cancel{e^{x^2}}(1 - 2x^2)}{\cancel{e^{x^4}}} = 0$$

e

$$1 - 2x^2 = 0$$

$$x = \pm\sqrt{\frac{1}{2}}$$

$$\sum_{n=1}^{\infty} \frac{n}{e^{n^2}}$$

$$\int_1^{\infty} \frac{x}{e^{x^2}}$$

$$\int \frac{x}{e^{x^2}}\,dx \qquad u = x^2 \qquad du = x\,dx$$

$$\int \frac{1}{e^u}\,du$$

15. $\int e^{ax}dx = \frac{e^{ax}}{a} + c$

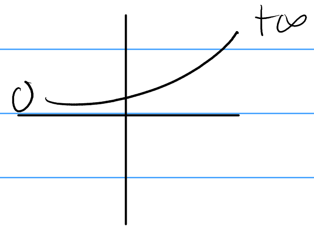
$-1 = a$

$$\int e^{-u}\,du \longrightarrow \int e^{-1u}\,du$$

$$\frac{e^{-u}}{-1} = -e^{-u} = -e^{-x^2}$$

$$-e^{-x^2}\Big|_1^{+\infty}$$

$$\lim_{x\to+\infty} -e^{-x^2} - -e^{-1^2}$$



$$-\cancel{e^{-\infty}}^{\,0} + e^{-1}$$

$$\boxed{\sum_{n=1}^{\infty} \frac{n}{e^{n^2}} \text{ Converge}}$$

$$0 + \frac{1}{e} = \frac{1}{e}$$, k, is converge

CADENA DE TÉRMINOS DOMINANTES

Sean $k \in \mathbb{R}, a, p \in \mathbb{R}^+, a > 1$ entonces para n suficientemente grande se tiene que:

$$k \ll \ln n \ll n^p \ll a^n \ll n! \ll n^n$$

$$\lim_{x\to+\infty} \frac{f(x)}{g(x)}$$, $f(x) > g(x) = +\infty$, $f(x) < g(x) = 0$

$$\lim_{x\to+\infty} \frac{\ln(x)}{n^n} = 0$$

$x+3 = x$

$$\lim_{x\to+\infty} \frac{n^n}{\ln(x)} = +\infty$$

K sumadas $= 0$

K multiplicadas $= 1$

$x+3 = x \qquad 3x+3 = x$

Criterio de la razon

La serie $\sum_{n=p}^{\infty}$ Que tenga ! o ...

Converge si $\lim_{n\to+\infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$

Diverge si $\lim_{n\to+\infty} \left| \frac{a_{n+1}}{a_n} \right| > 1$ 0 ∞

!!!!

$$\sum_{n=3}^{\infty} \frac{(3n+5)!}{20n-7} \qquad \lim_{n\to+\infty} \frac{a_{n+1}}{a_n}$$

$$a_{n+1} = \frac{(3(n+1)+5)!}{20(n+1)-7}$$

$$\frac{(3n+3+5)!}{20n+20-7}$$

$$\frac{(3n+8)!}{20n+13}$$

$$\lim_{n\to+\infty} \frac{\frac{(3n+8)!}{20n+13}}{\frac{(3n+5)!}{20n-7}}$$

$$\lim_{n\to+\infty} \frac{(3n+8)!\,(20n-7)}{(3n+5)!\,(20n+13)}$$

$$\lim_{n\to+\infty} \frac{(3n+8)(3n+7)(3n+6)\cancel{(3n+5)!}(20n-7)}{\cancel{(3n+5)!}(20n+13)}$$

$$\lim_{n\to+\infty} \frac{(3n+8)(3n+7)(3n+6)(20n-7)}{(20n+13)}$$

$$K \cdot +\infty = +\infty$$

$$K + \infty = +\infty$$

$$\lim_{n\to+\infty} \frac{n \cdot n \cdot n \cdot \cancel{n}}{\cancel{n}}$$, cadena de dominancia

$$n^3 = +\infty^3 = +\infty$$

$\therefore$ Diverge

$$n! = n(n-1)(n-2)(n-3)\ldots!$$

$$(3n+8)! = (3n+8)(3n+8-1)(3n+8-2)(3n+8-3)!$$

$$(3n+8)(3n+7)(3n+6)(3n+5)!$$